Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/15/2026
PI Name:	Wu, Shu-Chieh		
Project Title:	Lessening Impact of Interface Inconsistency Through Goal-Directed Crew Operations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	San Jose State University Research Foundation Inc.		
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City:	San Jose	**State:**	CA
Zip Code:	95192	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 HERO 80JSC024NA001-FLAGSHIP, OMNIBUS, and NSCOR Crew Health: Topics in Human Health Countermeasures, Behavioral Performance, and Space Radiation and NASA Human Research Program Omnibus Opportunity
Start Date:	02/19/2025	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/25 per L. Yu/HFBP (Ed., 9/2/25). NOTE: Period of performance (POP) has changed per NASA JSC and NSSC documentation. The new POP is 02/19/2025-02/18/2028. The original POP was: 10/01/2024-09/30/2027.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McTigue, Kaitlin B.S. (NASA Ames Research Center) Panontin, Tina Ph.D. (San Jose State University Research Foundation Inc) Parisi, Megan M.S. (NASA Ames Research Center) Ulrich, Thomas Ph.D. (Idaho National Laboratory) Boring, Ronald Ph.D. (Idaho National Laboratory)		
Grant/Contract No.:	80NSSC25K7254		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Over the course of a nominal mission beyond low-Earth orbit (LEO), crew are expected to operate a variety of space vehicles, executing different phases of the mission, with each one possibly featuring an idiosyncratic interface design. It is as if crew have to master using Macs, Windows, and Linux machines all at the same time and prepare to switch between them on demand, a non-trivial feat even for system admins, not to mention how much more complex space vehicles are to operate and consequential when errors occur. The central objective of the proposal is to test the idea that impact of interface inconsistency on crew performance can be lessened by designing vehicle management interfaces to better support goal-directed crew operations. Specifically, we hypothesize that crew performance is more vulnerable to interface inconsistencies when crew operations are rule-dictated (procedure-following), and primarily involve locating and identifying menu items corresponding to procedures. When crew can manage vehicle systems in a more goal-directed (procedure-utilizing) manner like flight controllers, (e.g., determining vehicle status and planning courses of actions), we hypothesize that crew performance would be less vulnerable to interface inconsistencies. Goal-directed system management, we surmise, may be promoted by designing interface functionality to better match vehicle system management task needs and crew capability, effectively the essence of Task-Technology Fit (TTF). Our method includes analyzing current vehicle management procedures and interviewing flight controllers to determine what interface functionalities would support TTF in space vehicle system management and, based on the output of the analyses, empirically assessing how interface consistency and level of TTF interact to influence operational performance. Output of the proposed work will inform how to design interfaces that support goal-directed crew operations, which have the potential to simultaneously lessen the impact of interface inconsistency, reduce the need for training to use specific interfaces, and enable the crew to perform increasingly autonomously amid reduced ground support due to communication delays.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research program and associated deliverables will contribute to reducing the Earth Independent Human-Systems Operations (EIHSO) risk, formerly known as Human-Systems Integration Architecture (HSIA) risk. They will also inform gaps in knowledge relevant to maintaining crew health and performance on future long duration exploration missions beyond low-Earth orbit (LEO).
Task Progress:	The project ended on September 30, 2025 after being terminated by the HRP.
Bibliography Type:	Description: (Last Updated: 06/24/2025)